

Atreides Archive Intelligence Assistant

AI Innovation SLIIT Codefest

Submission report

Team Atreides

Repository: [Atreides-AI-Innovation-Codefest](#)

Live deployment: [Atreides Archive](#)

1. Problem and chosen track

Enterprise and fictional archives are difficult to use because the answer to one question may be distributed across narrative text, structured tables, scanned pages, figures, and multiple documents that disagree. A conventional keyword search returns passages but does not know whether the retrieved material completes the question, how facts connect, whether two sources conflict, or when investigation should stop. The Ashen Era Archive makes this problem deliberate: it contains invented names, mixed document formats, cross-document relationships, unreliable in-world sources, and facts that exist only in artwork.

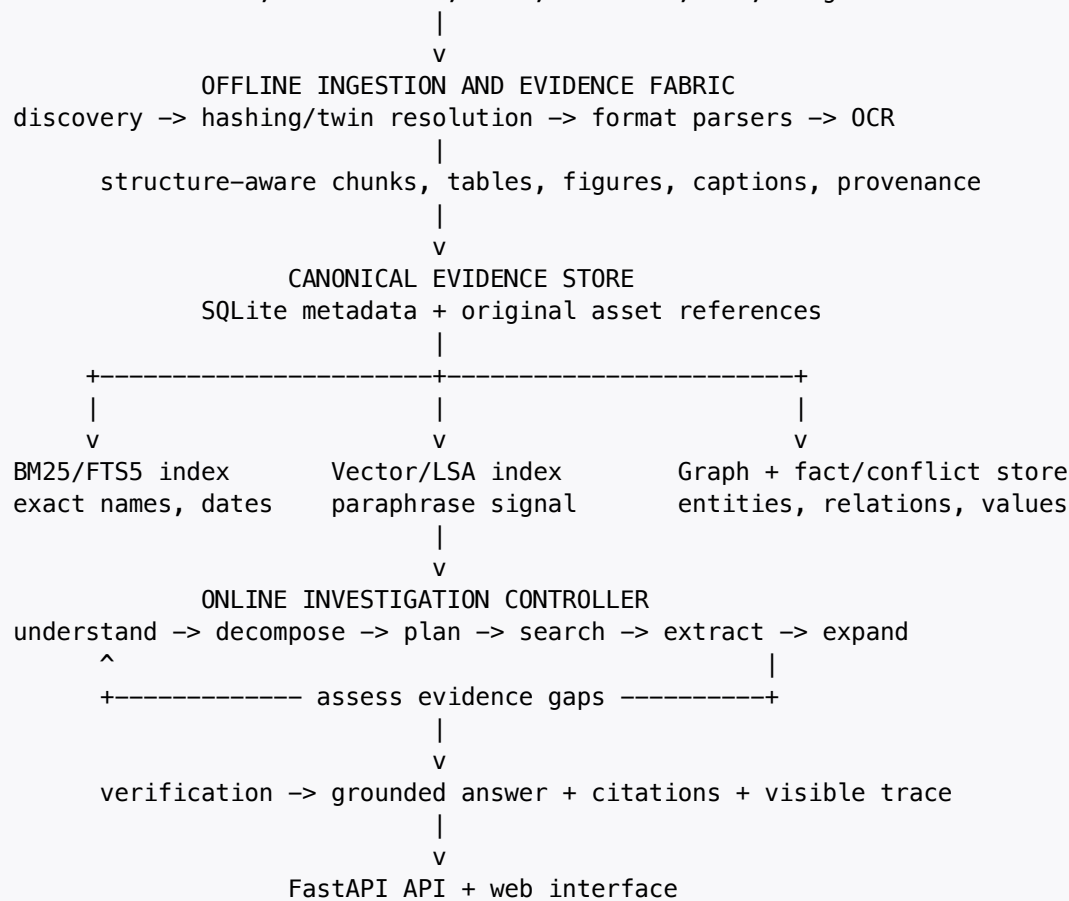
The selected primary sub-track is **1C - Searching the Way a Human Does**. The system treats a question as a bounded investigation: understand the request, decompose it into evidence requirements, search, extract, expand through newly discovered entities, check gaps, verify claims, and stop only with an explicit reason. The supporting sub-track is **1B - Connecting Facts Across Thousands of Pages**, implemented through provenance-carrying facts and graph traversal. Selected 1A-style visual evidence is also supported through OCR, chart reading, figure retrieval, and honest partial-visual responses. This positioning is consistent with the solution plan and the implemented repository scope.^[1]

The product is therefore not a general web-search engine or an ordinary database lookup. The archive remains the source of truth; indexes are navigation aids. Every answerable claim must point to an evidence chunk, document, page or asset, and the final interface exposes the evidence chain, confidence, conflicts, trace, and stop reason.

2. Solution and architecture

The solution is a provenance-first archive investigator with an offline indexing plane and an online investigation plane. The separation keeps ingestion deterministic and re-runnable while keeping query-time investigations stateless, bounded, and diagnosable.

Archive files: PDF, scanned PDF, DOCX, Markdown, TXT, images



Ingestion and evidence

The pipeline handles PDF, scanned PDF, DOCX, Markdown, plain text, and images. Parsers own provenance while the format is still known; downstream chunking may inherit page, section, and line metadata but cannot invent it. Tables and figures remain atomic rather than being split by fixed windows, preventing orphaned values such as a cost separated from the entity heading. Duplicate format twins are resolved so the same work is not indexed twice; the PDF is preferred where page citations are available.

The current full-archive build contains **339 discovered files, 321 indexed files, 18 format twins, 1,330 pages, 2,547 chunks, 227 tables, and 187 figures**. The derived knowledge fabric contains **417 entities, 1,912 provenance-carrying edges, 1,499 facts, and 34 normalized conflicts**.

Retrieval and investigation

Four complementary signals are used: exact BM25/FTS5 retrieval for invented names, dates, and titles; local LSA vectors for paraphrases; graph traversal for multi-hop relationships; and a structured fact/conflict store for precise attributes and competing values. Hybrid ranking also includes entity overlap and source reliability. The measured production split is lexical-heavy because the archive contains rare fictional vocabulary: the paraphrase probe selected a 0.50 lexical / 0.30 semantic balance, outperforming the plan's more vector-heavy proposal at recall@1 and recall@5.

The investigation loop uses explicit completion conditions. A multi-hop request such as “Who rules the place where the Gravemaw Wyrms live?” first retrieves the creature's lair, then uses the discovered place as the next query target, and finally verifies the ruler. A passage retrieval without a citable value is marked attempted but failed, never satisfied. Stop states include complete, insufficient evidence, no new evidence, no entity, budget exhaustion, and partial visual evidence.

Verification and interface

Claims are classified as **direct**, **inferred**, **conflicting**, **unsupported**, or **partial visual**. Direct claims quote a source; inferred claims show the supporting chain; conflicts preserve competing values and source reliability; unsupported claims are excluded from the factual answer. Confidence is structurally bounded by source count, reliability, hop count, OCR or vision confidence, and unresolved conflict penalties.

The FastAPI service and web UI show the answer first, followed by cited claims, quoted evidence, source reliability, evidence chain, graph path, conflicts, satisfied and failed sub-questions, investigation timeline, fallback events, and the exact stop reason. Asset routes resolve indexed IDs rather than arbitrary paths, protecting the archive from path traversal.

3. Key technical decisions

Decision	Rationale and trade-off
SQLite + FTS5 rather than a service-heavy stack	A judge can run the archive locally without OpenSearch, Neo4j, or a hosted vector service. This sacrifices distributed scaling, which is unnecessary for 1,330 pages, but improves reproducibility.
Provenance-first evidence objects	Citations are generated from recorded metadata, not invented by an LLM. Every fact and graph edge retains the chunk that asserted it.
Reliability as a ranking signal, not a filter	Lower-reliability ballads and wiki sources remain visible because they may expose the conflict being asked about. Codex reliability influences ranking and confidence without deleting evidence.
Deterministic extraction before LLM assistance	Explicit table rows and links provide a zero-API-cost baseline. LLM output must earn acceptance through archive entity matching, schema validation, and evidence support checks.
Bounded state machine instead of an autonomous agent	Hard limits on iterations, queries, graph hops, time, and stale rounds make live demonstrations reproducible and rate-limit safe. Every incomplete result reports why it stopped.
Local LSA before hosted embeddings	Hosted bulk embeddings were rate-limited on the tested free tier. Local LSA builds the 2,547-chunk index in about seven seconds, is deterministic and offline, and avoids a live-demo dependency.

One AI gateway	All provider calls pass through one boundary for schema validation, caching, retries, backoff, timeouts, circuit breaking, usage accounting, and deterministic fallback.
Never promote visual observations to facts	Vision can be wrong and the configured text model cannot see images. Visual outputs remain named observations with confidence and are not written into the permanent fact store.

4. Build status, failures, and limitations

What works

The complete path currently runs from ingestion through four indexes, hybrid retrieval, bounded investigation, reasoning, verification, API, and UI. The deterministic path works without an API key; optional LLM assistance adds query interpretation, gap-directed query suggestions, and candidate extraction but cannot overrule indexed entities, attributes, relations, or evidence. The repository records **30/30 on a held-out capability set, 30/30 on the team regression set, and 15/20 on the supplied cited development set**. The primary 1C set is reported as **2/2**, while the remaining supplied misses are mainly visual-artwork gaps.

The system supports direct lookup, inverse lookup, multi-hop traversal, multi-fact decomposition, comparison, superlative and ranking questions, duration, date ordering, date difference, existence checks, percentage, ratio, difference, total, average, conflict detection, reliability ranking, and insufficient-evidence handling. OCR covers the scanned PDFs and figure plates, while chart plates can be read geometrically when subject and reference bars are identifiable.

What does not work completely

The major remaining limitation is pictorial understanding. Heraldry, portraits, and paintings often contain no readable label, so OCR returns noise. If no vision model is configured or the provider is unavailable, the system names the relevant asset and returns `partial_visual` rather than guessing. A separate vision model can work intermittently, but image size and the shared free-tier daily cap make it unsuitable as the only path for a live demo.

Other boundaries are explicit: three-plus-hop questions are not fully recursive; causal and impact questions are detected but not separately reasoned; timeline output lists dated events without narrating a complete chronology; attribute routing is a maintained policy; multi-line table cells may be truncated by the current fact extractor; PDF section paths are heuristic; OCR errors are recorded but not yet repaired with fuzzy aliases; and graph completeness depends on explicit links and tables rather than all relations stated in prose.

Failed approaches and corrections

Several approaches were tested and rejected because they produced plausible but unsafe results:

- Marking a question complete when passages were retrieved caused “supported” answers with no citable value. Completion now requires evidence satisfying each sub-question.
- Reading hop direction from English grammar answered inverse questions backwards. Hop direction is now derived from the relation recorded by the named subject.
- Extracting tables before page text detached codex tables from the headings that identified their subjects. Text is emitted first and bound to the table.
- Requiring only wiki links for graph edges excluded authoritative codex relations written as plain values. Name-shaped plain values are now accepted, increasing typed edges from 110 to 218 during development.

- Generic “language-like” OCR checks surfaced convincing artwork garbage as evidence. Plates are now quotable only when recognized labels identify what the text describes.
- The proposed vector-heavy fusion weights were not assumed to be correct. A paraphrase probe showed that lexical weight must remain dominant for invented names, so measured weights replaced the original proposal.

5. AI usage disclosure

AI assistance used in implementation was **Claude Code using Claude Opus 5** for Phase 1 implementation, test authoring, documentation drafting, and interactive corpus investigation. The team—not the assistant—selected the sub-tracks, architecture, provenance rule, reliability policy, bounded state-machine design, build order, and decision trade-offs. The repository includes an AI usage disclosure and exported chat-log location.

AI-generated or AI-assisted code was reviewed against the real archive, tests, source pages, and evaluation results. The assistant was used to implement parsers, chunking, storage, graph and fact extraction, gateway logic, tests, UI, and documentation; it was not allowed to create authoritative citations or inject unsupported archive facts. The gateway rejects unknown entities, unsupported relations, invalid evidence IDs, and claims whose subject or value is not present in the supplied evidence. The design therefore treats AI as an optional interpreter and accelerator, never as the archive's source of truth.

The project also documents concrete AI errors that were corrected: table-before-text parsing, wikilink-only graph extraction, nearest-heading chunk binding, an incorrectly specified ranking test, and case-only false conflicts. These corrections were discovered through source inspection, evaluation, and regression tests rather than accepted from generated output.

6. Team members and contributions

Repository history verifies two named GitHub contributors. The available solution-plan and repository materials define four responsibility areas but do not provide a complete formal roster beyond the contributors below; additional team names should be inserted before final submission if required by the competition form.

Member	Verified role and contribution
Chanuka Janandhana	Lead implementation and integration: ingestion parsers, page-aware provenance, OCR, chunking, SQLite/FTS5, graph and fact extraction, retrieval, investigation loop, verification, AI gateway, API/UI integration, evaluation, documentation, and deployment-facing hardening. The repository history attributes the majority of core architecture and implementation commits to this account.
Randil Lasith	Collaboration and deployment contribution: VPS/automatic-deployment verification, investigation-iteration configuration, evidence-grounded comparison synthesis, and related integration work. These contributions are visible in the repository commit history under the randillasith account.
Nadeeshan Sameera, Piyanga Tharindu	The technical plan defines the intended shared responsibility areas—ingestion/provenance, retrieval/indexing, AI investigation, and verification/product—but no additional names were recoverable from the checked repository materials. Final submission metadata should replace this line with the remaining members and their confirmed contributions.[^1]

The implementation is designed so that all members can explain the complete evidence lifecycle: ingest and preserve provenance, retrieve through complementary indexes, investigate with bounded planning, verify claims, expose uncertainty, and demonstrate deterministic fallback behavior. This directly supports the challenge requirement that the solution be explainable and modifiable by the team rather than being a one-shot generated answer.

Submission links

- Repository: github.com/CJChANu/Atreides-AI-Innovation-Codefest
- Live prototype: atradias.duckdns.org
- Architecture and decisions: [docs/architecture.md](#), [docs/decisions.md](#)
- Limitations: [docs/limitations.md](#)
- AI disclosure: [ai_usage/ai-usage-disclosure.md](#)

References

1. [Final-Consolidated-Technical-Solution-Plan-Archive-Intelligence-Assistant.pdf](#) - The proposed solution is a provenance-first, multimodal archive intelligence assistant for the Ashen...